

References: Read Reactor Theory Manual Chapter 1: §1.4**On Moodle:**

- *K-Effective, Reactivity & Criticality Handout PDF*
- *Lab #3 – Normal Reactor Startup PDF*

1. **3 PTS:** Give 3 examples of nuclear reactions yielding “Source Neutrons” in the reactor core:

- Photo-neutrons from gammas on Beryllium (reflectors)
- Photo-neutrons from gammas on Deuterium (in water)
- ^{18}O (α , n) in fuel
- Spontaneous Fission
- Startup Source PuBe (α , n)

2. **6 PTS (0.5 pts each entry below):** List 2 “SOURCES” and 4 “SINKS” of **fission** neutrons (**not** source neutrons) in the reactor core, AND give the factor in the 6-FACTOR FORMULA that accounts for each:

SOURCES:		“6-Factor”
A.	Thermal Fission (prompt and delayed)	η
B.	Fast Fission	ϵ
SINKS:		
A.	Fast Leakage	$\mathcal{L}_f$
B.	Resonance Absorption	p
C.	Thermal Leakage	$\mathcal{L}_{th}$
D.	Non-Fuel Thermal Absorption	f

3. **5 PTS:** 6 Factor Formula Definitions - *MATCHING*

<u>Factor:</u>		<u>Definitions:</u>
<u>F</u>	η	A. Ratio of number of fast neutrons that start to slow down (moderate), to the number of fast neutrons from fast and thermal fission.
<u>C</u>	ϵ	B. Ratio of the number of thermal neutrons that are absorbed in the core, to the number of neutrons that reach thermal energy.
<u>D</u>	p	C. Ratio of fast neutrons from both fast & thermal fission, to the number of fast neutrons from thermal fission alone.
<u>E</u>	f	D. Ratio of the number of neutrons that reach thermal energy, to the number of neutrons that started to slow down (moderate).
<u>B</u>	$\mathcal{L}_{TH}$	E. Ratio of the number of thermal neutrons that are absorbed by the fuel, to the number of thermal neutrons that are absorbed in the core.
<u>A</u>	$\mathcal{L}_F$	F. Ratio of the number of neutrons generated from thermal fission, to the number of thermal neutrons absorbed in the fuel.

4. **5 PTS:** Fill in the blanks with either a value or a range of values where appropriate (i.e. $0 < \rho/\beta \leq \$0.5$ etc...) using the correct units. (Hint: if $\rho = \$1 = 730$ pcm then $K_{eff} = 1.0073$)

If the reactor is:	K_{eff} is:	Reactivity ρ/β (\$) is:
Super-prompt critical:	> 1.0073	$\rho > \$1$ (or 730 pcm)
Prompt-critical:	$= 1.0073$	$\rho = \$1$ (or 730 pcm)
Super delayed critical:	$1 < K_{eff} < 1.0073$	$0 < \rho < \$1$ (or 730 pcm)
Critical:	$= 1$	$= 0$
Sub-critical:	< 1	< 0

Lab Questions

REF: Reactor Startup Lab Handout

- 3 PTS:** What are the three characteristics of subcritical multiplication that were observed while taking the reactor from shutdown to (super) critical?
 - Doubling the Source Range Monitor (SRM) count-rate halves the difference between the current value for K_{eff} and 1, therefore taking the reactor halfway to criticality.*
 - Withdraw the control rods and double the SRM count-rate approximately 5 times to bring the reactor to criticality (or just super-critical).*
 - The closer you get to criticality, the longer it takes for the SRM count rate to reach subcritical equilibrium (i.e. a stable non-increasing neutron population).*
- 2 PTS:** What are two indications on the reactor console that the reactor is super-critical?
 - Linearly increasing traces on the SRM and Log-N semi-log plot recorder traces (without control rod motion) indicates an exponential increase in the reactor neutron population. This is an indication that the reactor is **SUPER-CRITICAL**.*
 - A **Positive Stable Period (τ)** (without control rod motion) as indicated on the SRM and Log-N Channel meters confirms an exponential increase in neutron population. This is another indication that the reactor is **SUPER-CRITICAL**.*
- 1 PTS:** Which indicates a faster rate of power increase, a 10 second or a 30 second Period τ (sec)?

A 10 second period. The period is the time it takes in seconds for the power to change by a factor of 'e' (2.718), so a shorter period means the power is changing more quickly.
- 1 PTS:** According to INPO, what is the first thing you should do before performing a reactor operations procedure?

STOP & THINK – what operations evolution/procedure is being performed?